Fiscal Year:	FY 2017	**Task Last Updated:**	FY 08/11/2016
PI Name:	Mao, Xiao Wen M.D.		
Project Title:	Space Flight Environment Induces Remodeling of Vascular Network and Glia-Vascular Communication in Mouse Retina		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	92350-0001	**Congressional District:**	31
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	11/01/2014	**End Date:**	04/30/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 4/30/2019 per F. Hernandez/ARC (Ed., 6/27/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gridley, Daila Ph.D. (Loma Linda University) Pecaut, Michael Ph.D. (Loma Linda University) Sun, Shu-Wei Ph.D. (Loma Linda University)		
Grant/Contract No.:	NNX15AB41G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The eye is a unique organ because it is relatively unprotected and constantly exposed to atmospheric oxygen, background radiation, environmental chemicals, and physical abrasion. A recent report shows that more than 30% of the astronauts returning from space were diagnosed with eye problems that can cause headaches and blurry vision. Our Preliminary studies from mice that had been subjected to space flights showed that environmental conditions during space travel lead to oxidative stress and induce adverse microvessel remodeling in the retina. However, our knowledge about the pathophysiological process from adaptive response to irreversible oxidative damage in the retina during and after space flight exposure and the underlying cellular mechanism(s) of these disturbances is very limited and inconsistent. The objectives of this application are to characterize the impact of space flight on female mouse retinal vasculature and tissue remodeling and to study the role of mitochondria in regulating oxidative stress-induced cellular damage and cell-cell interactions in the function of blood-retina-barrier (BRB). Our study will also determine the extent to which an antioxidant metalloporphyrin protects against space flight environment-induced changes in retinal neurovascular coupling and retinal function. The hypothesis will be tested in the specific aims with the spaceflight experiments on the International Space Station (ISS) and compared to that of ground-based groups using matured female mice. For flight studies, there will be two experiments over two flights, one for in-flight adaptation study on the ISS and one for studying re-adaptation after live animal return to Earth following long-term ISS mission. Selected groups of mice will be treated intraperitoneally with an antioxidant a week before launch and weekly during space flight. Animals will be sacrificed in orbit at 14 and 30 days after launch or 30 days and up to 6 months after return alive on Earth. Together, our unique, integrative, and quantitative activities with advanced MRI (magnetic resonance imaging) imaging techniques and retinal functional testing using visual electroretinography will provide insight into the molecular mechanisms and pathways of space flight condition-induced mitochondrial oxidative damage on retinal BRB integrity, vascular remodeling, and retinal function. The data will provide unique criteria for risks of functional detriments associated with space flight. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward potential countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward potential countermeasures.
Task Progress:	Because our flight experiment has not been scheduled, we were provided the option of expanding our Definition Phase Research in order to initiate our studies. Definition phase of the study was just approved and funded in April. To facilitate collaboration with other funded Space Biology investigators, enhancing scientific productivity, Dr. Jeff Willey and I just started collaborative study to examine the combined effects of radiation exposure and unloading to retina. We will use 80 female C57BL/6 mice for the Definition Phase of Specific Aim 1) to define the relationships between space flight condition-induced oxidative stress in ROS expression, retinal vascular remodeling, and BRB function; and 2) to determine whether space flight condition-induced oxidative damage in retina is mediated through photoreceptor mitochondrial ROS production. Each group will include 10 mice. A group of 40 mice will receive hindlimb unloading (HLU) using our tail suspension apparati. An additional 40 mice will serve as full-weight bearing ambulatory controls. On Day 5, half of the mice from each group will be irradiated during tail suspension without anesthesia using our published procedure to simulate space flight radiation exposure during periods of HLU. The high-energy radiation will be delivered with a Varian 2100 SC medical LINAC at an accelerating potential of 10 MV, which will minimize the depth-dependent dose fall off during exposure. The dose rate of ~1 Gy/min will be used to minimize dose heterogeneity. Mice will then remain either HLU or ambulatory for the next 25 days, at which time the mice will be sacrificed. Left eye will fixed in 4% PFA and right eye will be frozen for analysis. Approved Specific Aim, procedures, assays, and outcomes will be identical to the original proposal. Since this is an ongoing study, we currently have no data to present. An antioxidant metalloporphyrin will be used in this study to determine if it can protect against space flight environment-induced changes in retinal neurovascular coupling and retinal function. To gain insight into the tissue distribution as well as to provide data that would ensure proper dosing regimens, we will conduct ground-based comprehensive pharmacokinetic study. Collaborators have been contacted to discuss the experiment design for the ground-based study.
Bibliography Type:	Description: (Last Updated: 07/10/2026)
Abstracts for Journals and Proceedings	Mao XW, Crapo J, Gridley DS. "Radioprotection of Rat Eyes by a Superoxide Dismutase Mimic: a Metalloporphyrin Compound." Presented at 17th International Conference on Antioxidants. Paris, France, June 13-15. 2016. Journal of International Society of Antioxidants in Nutrition & Health. 2016 Jun;3(3):Special Issue for Paris Redox 2016. , Jun-2016